

NetSecure Project Structure & Team Roles

This document explains the recommended project structure for the NetSecure cybersecurity project. It also defines which team member will work in each directory and what responsibilities they will handle.

Recommended Project Structure

```
network-scanner/
├── scanner/                # Person 1
│   ├── scanner.py
│   ├── ports.py
│   ├── services.py
│   └── risk_detection.py
├── backend/               # Person 2
│   ├── app.py
│   ├── routes.py
│   ├── database.py
│   └── models.py
├── frontend/             # Person 3
│   ├── templates/
│   ├── static/
│   │   ├── css/
│   │   ├── js/
│   │   └── charts/
├── reports/
├── tests/
├── requirements.txt
├── main.py
└── README.md
```

Person 1 — Scanner Engineer + Project Lead

Main Directory: scanner/

Files They Work On:

- scanner.py
- ports.py
- services.py
- risk_detection.py

Main Responsibilities:

- Build the port scanner
- Detect open ports
- Detect running services
- Create risk detection logic
- Improve scan speed and threading
- Coordinate project integration

Main Goal:

Create the cybersecurity scanning engine of the project.

Person 2 — Backend + Database Engineer

Main Directory: backend/

Files They Work On:

- app.py
- routes.py
- database.py
- models.py

Main Responsibilities:

- Build Flask backend APIs
- Connect scanner with frontend
- Store scan history in database
- Manage application routes
- Handle backend integration logic

Main Goal:

Build the backend architecture and data management system.

Person 3 — Frontend + Dashboard Engineer

Main Directory: frontend/

Files/Folders They Work On:

- templates/
- static/css/
- static/js/
- static/charts/

Main Responsibilities:

- Build the dashboard UI
- Create scan forms
- Display results visually
- Create charts and graphs
- Improve project appearance and user experience
- Create README screenshots and visual polish

Main Goal:

Create a professional and user-friendly interface.

Shared Directories & Files

Directory/File	Purpose	Who Uses It
reports/	Stores scan reports and exported files	All Team Members
tests/	Testing scanner and application features	All Team Members
requirements.txt	Stores project dependencies	All Team Members
main.py	Main project entry point	Project Lead + Backend Engineer
README.md	Project documentation and setup guide	All Team Members

Team Workflow

1. Each person works in their own branch.
2. Code is pushed regularly to GitHub.
3. Features are tested before merging.
4. Stable features are merged into the main branch.
5. Everyone understands the full project flow even with separate roles.